

PROJECT TITLE :

Retail Sales Intelligence : Statistical Analysis of Store Performance

Project Overview

In today's competitive retail landscape, data-driven decision making is essential for business growth. This project demonstrates how statistical analysis can transform raw sales data into actionable insights for inventory optimization and marketing strategy.

Business Context

Retail managers constantly face challenges in understanding product performance across different categories. Without proper analytics, it's difficult to identify which products drive revenue, maintain optimal stock levels, or allocate marketing budgets effectively.

Problem Statement

A retail store experienced inconsistent sales performance across product categories. Management needed to:

1. Identify underlying sales trends and patterns
2. Validate whether average sales volumes meet business targets (20 units benchmark)
3. Compare performance across Electronics, Clothing, Home, and Sports categories

Project Objectives

- Descriptive Analytics: Summarize sales performance using statistical measures
- Inferential Analytics: Test hypotheses about sales targets using t-tests and confidence intervals
- Data Visualization: Create intuitive charts for stakeholder decision-making

Dataset Description

I created a synthetic dataset simulating real-world retail conditions:

Feature	Description
product_id	Unique identifier for 20 products
product_name	Product description
category	Electronics, Clothing, Home, or Sports
units_sold	Daily sales volume (Poisson distribution)
sale_date	Transaction timestamp

Analytical Approach

Descriptive Statistics

I calculated fundamental metrics to understand sales distribution:

- Central Tendency: Mean, median, and mode of units sold

- Variability: Variance and standard deviation to measure consistency
- Category Breakdown: Total and average sales by product category

Inferential Statistics

- Confidence Intervals: Estimated population mean range to assess sales consistency
- One-Sample t-test: Tested whether mean sales significantly differ from the 20-unit target ($H_0: \mu = 20$ vs $H_1: \mu \neq 20$)

Key Visualizations

Chart Type	Purpose
Histogram	Distribution of sales volume with mean/median/mode markers
Box Plot	Category-wise spread and outlier detection
Bar Chart	Comparative total sales by category

Technical Implementation

- Language: Python (Pandas, NumPy, SciPy, Matplotlib/Seaborn)
- Statistical Methods: Parametric testing, distribution analysis
- Data Generation: Poisson distribution for realistic count data simulation

Results Summary

This analysis provided concrete insights into sales performance patterns. The statistical tests revealed whether the store meets its 20-unit target, while visualizations highlighted category-specific opportunities for inventory adjustment.

Key Takeaways

- Statistical rigor separates assumptions from facts in business decisions
- Synthetic data generation allows method testing before production deployment
- Clear visualizations bridge the gap between analysis and stakeholder understanding
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Skills Demonstrated

- ✓ Statistical hypothesis testing
- ✓ Data visualization & storytelling
- ✓ Python data analysis workflow
- ✓ Business problem translation to technical solutions